

HOMING AND ORIENTATION IN THE SPOTTED SPINY LOBSTER, *PANULIRUS GUTTATUS* (DECAPODA, PALINURIDAE), TOWARDS A SUBTIDAL CORAL REEF HABITAT

BY

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ABSTRACT

Panulirus guttatus (Latreille, 1804) is a sedentary, non-migratory species of spiny lobster that lives in the coral reef habitat throughout its benthic life. Homing and orientation of *P. guttatus* were investigated through experimental displacements of individual lobsters. Lobsters were caught in fixed traps deployed around three coral patches along a reef tract in northern Quintana Roo, Mexico. A total of 145 adult lobsters were tagged and individually released in one of four compass directions corresponding to the following sites as related to their home reef patches: in the reef lagoon (310°, ~W), along the axis of the reef tract (20°, ~N, or 200°, ~S) and in the fore-reef (100°, ~E). In each direction, individuals were displaced over 50, 100, or 200 m away from their point of capture. Thirty lobsters were recaptured, 29 of which in the same area of the reef patch where they had been initially captured, regardless of the time elapsed (1-146 days). The proportion of recaptured lobsters was independent of the direction and distance of release when lobsters were displaced 50 and 100 m into the reef lagoon or along the axis of the reef, but no lobsters were recaptured from a distance of 200 m along the axis of the reef. Hence, the familiar home range of *P. guttatus* appears to lie within a radius of 100 m along the reef tract, and within this home range individuals possibly use several crevices as refuges. A tethering experiment showed that lobsters significantly moved towards the reef when released on bare sand 500 m away from the reef. Wave surge may have oriented the lobsters towards the reef. These results indicate that, despite their sedentary, non-migratory nature, adult male and female *P. guttatus* show homing and orientation abilities.

RESUMEN

Panulirus guttatus (Latreille, 1804) es una especie de langosta sedentaria, no migratoria, que vive en el hábitat arrecifal durante toda su vida béntica. Se investigó la habilidad de *P. guttatus* para regresar al refugio y para orientarse por medio del desplazamiento experimental de langostas individuales. Las langostas se capturaron en trampas fijas colocadas alrededor de tres parches coralinos a lo largo del tracto arrecifal en el norte de Quintana Roo, México. Un total de 145

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langostas adultas fueron marcadas y liberadas en una de cuatro direcciones en relación con sus parches arrecifales originales: en la laguna arrecifal (310° , $\sim W$), a lo largo del eje del tracto arrecifal (20° , $\sim N$, o 200° , $\sim S$), y en el arrecife frontal (100° , $\sim E$). En cada dirección, las langostas fueron desplazadas 50, 100 o 200 m de su punto de captura. Se recapturaron 30 langostas, de las cuales 29 se encontraron en la misma área del parche arrecifal donde originalmente se capturaron, independientemente del tiempo transcurrido (1-146 días). La proporción de langostas recapturadas fue independiente de la dirección y distancia de liberación en individuos desplazados 50 o 100 m en la laguna arrecifal o a lo largo del tracto arrecifal, pero no se recapturaron individuos desplazados 200 m a lo largo del eje del tracto arrecifal. Por tanto, el ámbito hogareño familiar de *P. guttatus* parece comprender un radio de 100 m a lo largo del tracto arrecifal, y dentro de este ámbito hogareño los individuos pueden utilizar varios refugios. Un experimento con langostas atadas mostró que las langostas tendieron significativamente a moverse hacia el arrecife coralino después de ser liberadas en un fondo arenoso a 500 m del arrecife. La dirección del oleaje puede haber orientado a las langostas hacia el arrecife. Los resultados indican que, a pesar de su naturaleza sedentaria y no migratoria, los machos y hembras adultos de *P. guttatus* muestran habilidades para orientarse y regresar a su ámbito hogareño.

INTRODUCTION

Benthic decapods depend for their survival on the availability of suitable shelters and on their ability to find these shelters (Vannini & Cannicci, 1995). On hard substrates, such as coral reefs, large crevices are less frequent than small crevices, and their scarcity can adversely affect the population growth of large decapod species such as spiny lobsters (Palinuridae) (cf. Caddy, 1986). Therefore, mobile decapods must balance the need to find and defend a suitable crevice to avoid predation, with the opposite need to leave the shelter to forage or to seek mates (Vannini & Cannicci, 1995). Hence, many decapods have developed the ability to relocate their refuges; this ability is known as "homing" (Herrnkind, 1980). Homing was defined by Papi (1992) as any movement undertaken to reach a spatially restricted area that is known to an animal. Decapods may use, among others, visual, hydrodynamic, magnetic, topographic, and/or chemical cues to orient themselves in their home range, and to return to their home range after active or passive displacements (review in Vannini & Cannicci, 1995).

The spotted spiny lobster, *Panulirus guttatus* (Latreille, 1804) and the Caribbean spiny lobster, *P. argus* (Latreille, 1804) co-occur throughout the Caribbean Sea, the Bahamas, and Bermuda (Holthuis, 1991). Both species exhibit gregarious social behaviour, wherein two or more individuals co-den. But whereas *P. argus* has several changes in habitat during its ontogeny and is a highly migratory species, *P. guttatus* is a sedentary, typically non-migratory species that dwells in the coral reef habitat throughout its entire benthic life (Herrnkind, 1983; Briones-Fourzán, 1995; Evans et al., 1995; Sharp et al., 1997). Moreover, individuals of *P. guttatus* are highly reclusive, and only leave their shelters for short periods during darkness (Lozano-Álvarez & Briones-Fourzán, 2001).

Movement patterns have been widely studied in migratory spiny lobsters, such as *Panulirus argus*, *P. cygnus* George, 1962, *P. ornatus* (Fabricius, 1798), and *Jasus edwardsii* (Hutton, 1875). All these species exhibit nomadic movements spanning several kilometers, as well as seasonal mass migrations of up to 500 km (review in Herrnkind, 1983). In *P. argus*, hydrodynamic and magnetic cues have been advocated as important in homing, or during mass migrations (Herrnkind & McLean, 1971; Herrnkind et al., 1975; Walton & Herrnkind, 1977; Herrnkind, 1980; Nevitt et al., 1995; Lohmann et al., 1995). However, whether sedentary, non-migratory species such as *P. guttatus* use similar cues remains unknown. Similarly, although the sedentary nature of *P. guttatus* points to an ability of individuals to recognize their familiar surroundings, the home range of individual *P. guttatus* has not been determined. Experimental displacements, in which individuals are removed from their position and released at different distances from their home reefs, can provide useful information on these issues (Chittleborough, 1974; Vannini & Cannicci, 1995).

We explored the home range as well as the homing and orientation abilities of individual *P. guttatus* through experimental displacements. Because *P. guttatus* is an obligate resident of coral reef habitats and is believed to remain in the vicinity of its home reef throughout its benthic life, we hypothesized that individuals of *P. guttatus* would have a relatively small home range and would not be able to relocate their home area if displaced over long distances from that area.

MATERIALS AND METHODS

Study site. — The study was conducted on the coral reef at Puerto Morelos (20°51'N 87°56'W), northern Quintana Roo, on the Mexican Caribbean coast (fig. 1a). Puerto Morelos is located in the northern portion of an extensive barrier-fringing reef tract that extends from Belize to the Yucatan Strait. Rather than a continuous barrier, the coral reef in Puerto Morelos consists of a series of reef patches, separated from the coast by a shallow (<5 m deep) reef lagoon of 300–1000 m width. The sloping fore-reef has relatively few high-relief features, but hard coral cover tends to be dense at the reef crest and in the back-reef zone (Ruiz-Rentería et al., 1998), providing an intricate habitat with numerous crevices and caves where *Panulirus guttatus* dwells.

Stage 1: homing. — The study site consisted of three large, contiguous coral reef patches along the reef tract, separated by smaller patches and/or calcareous pavement. In October 1982, thirty plastic lobster traps were deployed around the three large reef patches (fig. 1b). Distance between adjacent traps ranged from 10 to 20 m. The traps were baited with cowhide and positioned by SCUBA divers

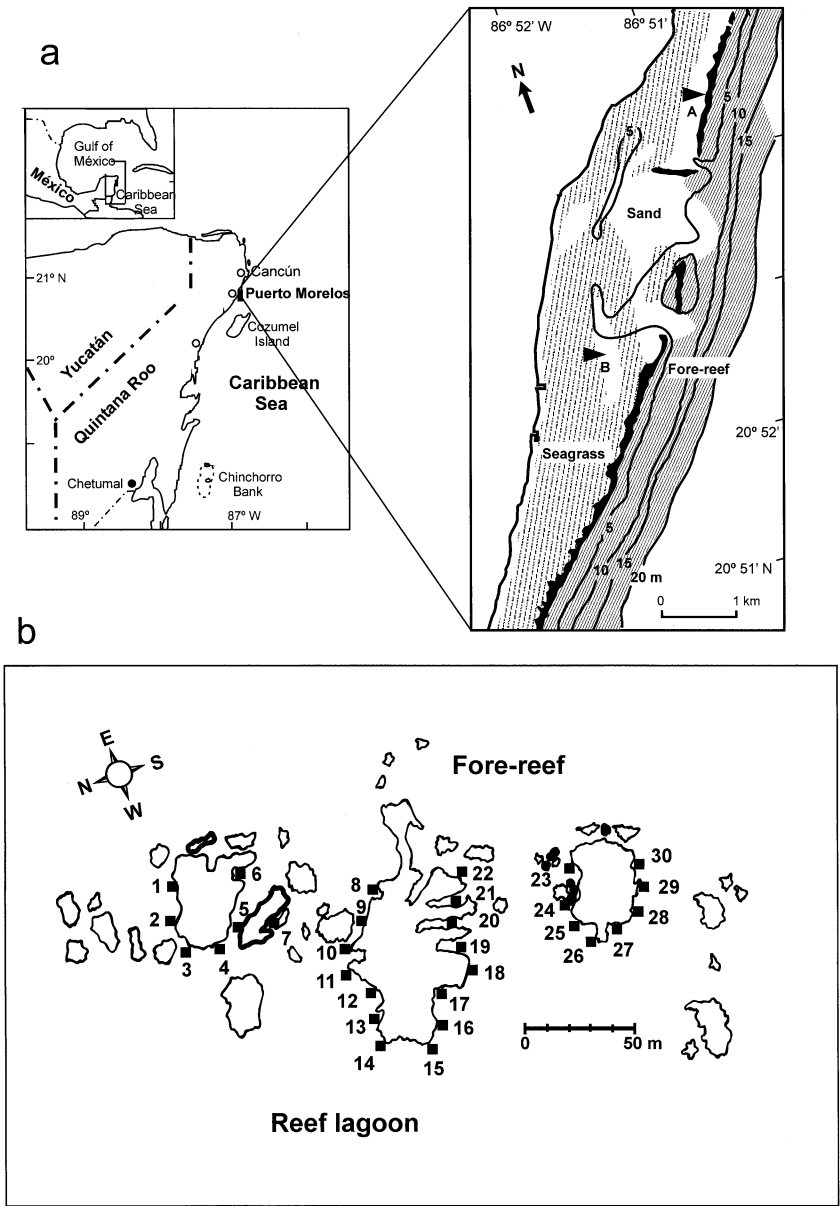


Fig. 1. a, Location of study area (coral reef at Puerto Morelos, Mexico); in the figure on the right, black areas denote the breaker zone of the reef tract; the black triangles A and B indicate, respectively, the study sites for homing and orientation; b, schematic representation of the reef patches along the reef tract where individual *Panulirus guttatus* (Latreille, 1804) were captured for the study on homing; black squares indicate the fixed position of each of the 30 traps deployed around the reef patches; the traps (not to scale) were positioned under ledges or as close to the reef structure as possible.

below ledges or coral formations in the back-reef zone. Each trap was individually numbered and remained fixed in the same place throughout every sampling period (Chittleborough, 1974). Divers checked the traps and extracted the lobsters daily for 15-20 days every month, during the dark part of the lunar cycle (third quarter to first quarter), because lobsters are more active on dark nights (Kanciruk, 1980). The bait was changed once a week.

Carapace length (CL, in mm) of lobsters was measured with calipers (± 0.1 mm). Each lobster was applied an individually numbered spaghetti-type tag (Floy Tag), inserted in the dorsolateral muscle between cephalothorax and abdomen. To minimize the possibility of tag loss, only individuals in intermoult (with a hard carapace) were tagged, and to ensure that recaptured lobsters would be identified even if they did lose their tag, holes were also punched in their telson or uropods following a code that represented the same number as the spaghetti tag. The number of the trap where each individual lobster was caught was recorded. Lobsters were transported in seawater containers by boat to the Puerto Morelos Academic Unit of the National University of Mexico, approximately 6 km from the study site, and kept in submerged traps under a pier until their time of release. Lobsters were released on the night following the day of capture, or in the afternoon of the following day.

Lobsters were randomly assigned to be released in one of four compass directions from their patches of original capture: in the reef lagoon (310° , $\sim W$), in the fore-reef (100° , $\sim E$), or along the axis of the reef tract (20° , $\sim N$, or 200° , $\sim S$) (fig. 1). Along these directions, individual lobsters were displaced either 50, 100, or 200 m from their original site of capture. Table I shows the number of lobsters released in each direction and distance. In the reef lagoon, lobsters were released at night (20:00-21:00 h), but in the remaining directions lobsters were released during the afternoon (13:00-16:00 h) because of navigational risks in the coral reef area at night. In the fore-reef, lobsters were not released at 50 and 100 m owing to the strong wave surge and turbulence caused by the breaking of waves on the reef. Each lobster was individually carried by a diver to its point of release, where it was dropped from the surface of the water to allow the lobster to sink freely to the bottom. When the lobster reached the bottom, the diver observed it for 7 min to record its general behaviour. Red lamps were used in night observations. Whenever an individual lobster was recaptured, the number of the trap was recorded to estimate its distance from the trap of original capture. We tested whether the proportion of recaptured lobsters was independent of the direction and distance of displacement with multiple contingency tables (Zar, 1984).

Stage 2: orientation. — In March-April 1985, 25 individual *P. guttatus* (size range: 54.0-75.5 mm CL) were caught to study their orientation abilities. For this purpose, we selected a sandy, featureless area in the reef lagoon, approximately 500 m from the coral reef and at a depth of 3.8 m. An iron stake, 30 cm long,

TABLE I

Capture and recapture data of spiny lobsters *Panulirus guttatus* (Latreille) displaced in different directions and at different distances from their point of initial capture; coral reef at Puerto Morelos, Mexico

Direction of release	Location of release relative to point of capture	Distance of displacement (m)	Number of lobsters released	Number of lobsters recaptured	Percentage of lobsters recaptured
310°	Reef lagoon (~W)	200	20	10	50.0
		100	15	2	13.3
		50	14	2	14.3
		Total:	49	14	28.6
200°	Axis of reef tract (~S)	200	11	0	0.0
		100	15	3	20.0
		50	12	2	16.7
		Total:	38	5	13.2
20°	Axis of reef tract (~N)	200	21	0	0.0
		100	15	6	40.0
		50	16	3	18.8
		Total:	52	9	17.3
100°	Fore-reef (~E)	200	6	2	33.3
		Total:	6	2	33.3

was fixed to the bottom, and a revolving PVC cylinder with a 30 m piece of monofilament line coiled to its exterior was secured to the stake. An individual *P. guttatus* was tethered to the loose end of the monofilament. The lobster was then dropped by a diver from the water surface above the stake, to allow the lobster to sink freely to the bottom. When the lobster had walked the full length of the tether and the monofilament was stretched taut, the angle was measured with a compass (Walton & Herrnkind, 1977). This procedure was repeated with 25 lobsters in all. The procedures were conducted at night and divers used red lamps to observe the lobsters from a distance of ~5 m. During each of these events, the direction of the waves, the wind, the prevailing current, and the ripples in the sandy bottom were measured with a compass. The distance between ripples was measured with a metal ruler, and current velocity with a surface current meter. The circular distribution of the angles of the lobster headings was analysed with a Rayleigh test (Zar, 1984), and the expectation of the lobsters to head towards the coral reef (110°) was tested with a V-test (Durand & Greenwood, 1958).

RESULTS

Homing

In total, 145 adult *Panulirus guttatus*, 97 males (size range: 55.0-82.3 mm CL) and 48 females (55.3-65.5 mm CL), were tagged and released. Six females were carrying external eggs when released. Thirty individuals (20.7%) were recaptured, 22 males and eight females, throughout the size range of those released. There were no significant differences in sex ratios of released and recaptured individuals ($\chi^2 = 0.474$, $df = 1$, $P > 0.25$).

Recaptures were less from along the axis of the reef than from the reef lagoon or fore-reef (table I). For example, of the 49 lobsters displaced into the reef lagoon (310°), 14 (28.6%) were recaptured, 10 of which had been displaced 200 m into the reef lagoon from their original reef patch. In contrast, only 14 (15.6%) of the 90 lobsters displaced along the axis of the coral reef (38 at 200° and 52 at 20°) were recaptured, all of them from 100 or 50 m, and two of the six lobsters (33.3%) displaced 200 m in the fore-reef (100°) were recaptured. Data from the fore-reef (100°) were not included in the contingency tables because we only released lobsters at 200 m in this direction. The proportion of recaptured lobsters was not independent from the directions (310°, 200°, and 20°) and distances (50, 100, or 200 m) of displacement ($\chi^2 = 25.952$, $df = 12$, $P < 0.025$). However, examination of the values in table I shows that the proportion of recaptured lobsters that were released at 200 m was highly different among the three directions tested. Thus, we suspected that the significant χ^2 was due largely to the data from 200 m. When the results from 200 m were excluded from the analysis (Zar, 1984), a nonsignificant χ^2 was obtained ($\chi^2 = 4.483$, $df = 7$, $P > 0.50$), indicating that the proportion of lobsters that were recaptured was independent of their release at 50 or 100 m along the axis of the reef tract (20° or 200°) or in the reef lagoon (310°).

After landing on the bottom, the lobsters displayed a wide range of movements and behaviours. In the reef lagoon, where lobsters were released at night, the distance moved by these lobsters (as measured in a straight line) during the 7 min of observation was 0-30 m, i.e., some lobsters remained still during the whole observation period whereas other started walking almost immediately. Most wandered at random, stopping and changing direction on numerous occasions. In contrast, along the axis of the coral reef and in the fore-reef, lobsters were released in daytime. These individuals also showed many types of behaviour, but the majority tended to rapidly seek shelter in any available crevice, where they remained throughout the observation period.

Lobsters remained at large from 1 to 146 days between subsequent captures, but the time elapsed apparently had no effect on their place of recapture. Except

TABLE II

Recapture data of individual *Panulirus guttatus* (Latreille) caught and displaced in different directions (20°, 100°, 200°, and 310°) and at different distances (50, 100, or 200 m) from their home patches. Coral reef at Puerto Morelos, Mexico (see fig. 1b for position of numbered traps). M, male; F, female; an asterisk indicates that the female was carrying external eggs when displaced

Lobster		Number of trap of		Distance between C and R traps (m)	Days at large	Direction of release	Distance of displacement (m)
Number	Sex	Capture (C)	Recapture (R)				
1	M	14	12	25	45	310°	200
2	M	16	14	30	2	310°	200
3	F*	19	18	10	103	310°	200
4	F	19	18	10	47	310°	200
5	F*	8	8	0	45	310°	200
6	M	10	13	30	103	310°	200
7	M	10	12	20	34	310°	200
8	M	11	8	40	146	310°	200
9	M	13	11	25	31	310°	200
10	M	11	13	25	63	310°	200
11	M	16	9	50	144	310°	100
12	M	12	13	10	56	310°	100
13	M	22	21	10	43	310°	50
14	M	13	11	25	70	310°	50
15	M	21	21	0	29	100°	200
16	M	21	19	20	35	100°	200
17	F	29	29	0	31	200°	100
18	M	23	27	35	17	200°	100
19	M	30	30	0	14	200°	100
20	M	25	26	10	62	200°	50
21	M	27	30	30	63	200°	50
22	M	3	3	0	16	20°	100
23	F*	3	3	0	23	20°	100
24	F*	3	2	15	20	20°	100
25	M	1	3	30	1	20°	100
26	M	7	3	40	14	20°	100
27	F*	3	2	15	5	20°	100
28	M	3	1	30	2	20°	50
29	F*	3	1	30	39	20°	50
30	M	2	25	185	41	20°	50

for lobster 30 (table II), which was recaptured on a different reef patch (~185 m away), all recaptured lobsters returned to their original reef patch, and 80% were found in the same trap or up to two traps (~30 m) away from the trap of original capture (table II; see fig. 1b for location of traps 1-30). This indicates that virtually all recaptured lobsters returned not only to the same reef patch, but to the same area in that particular patch where they had been initially caught.

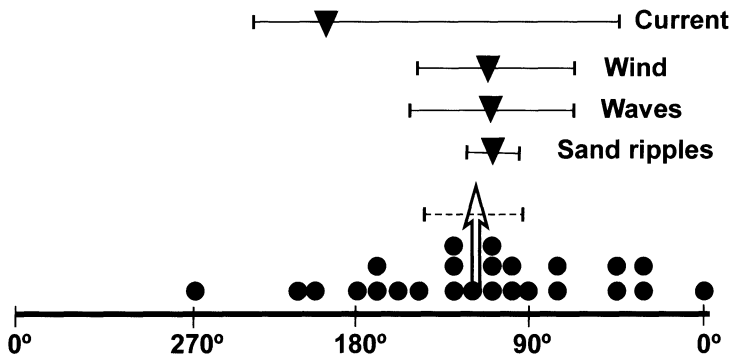


Fig. 2. Headings of 25 individual *Panulirus guttatus* (Latreille, 1804) (dots) displaced 500 m from the reef tract into the reef lagoon. The lobsters were tethered to a revolving cylinder containing 30 m of tether. The angles were measured after the lobsters walked the full length of the tether. The white arrow and the horizontal broken line indicate, respectively, the mean $\pm$ SD angle ($115.6 \pm 26.7^\circ$) of headings. The coral reef lies at 110° from the point where lobsters were tethered. The black arrowheads and their corresponding horizontal lines indicate the mean $\pm$ range of the direction of the water current, wind, waves, and sand ripples during the study period.

Orientation

The mean angle $\pm$ SD of the headings taken by the 25 tethered lobsters after walking 30 m was $115.6^\circ \pm 26.7^\circ$. The value of the mean vector ($r = 0.564$) indicated a relatively wide dispersion of the angles, but nevertheless the angles were not uniformly distributed around a circle (Rayleigh's test, $z = 7.9631$, $N = 25$, $P < 0.001$) (fig. 2). The V-test indicated a strong tendency of lobsters to head back towards the reef (110° , $u = 3.9717$, $P < 0.0005$). The bearing of the reef was very similar to the mean angle of the incoming wave surge (110.6° , range $75\text{--}150^\circ$), the wind (112.0° , range $75\text{--}145^\circ$), and the sand ripples (110.0° , range $105\text{--}120^\circ$) (fig. 2). The latter were separated by 42 cm on the average (range: 30–60 cm). In contrast, the surface current had an average direction of 217° (range $45\text{--}240^\circ$), and an average velocity of 7.1 cm s^{-1} (range: $4.1\text{--}9.9 \text{ cm s}^{-1}$).

DISCUSSION

In the time since the present study was conducted, movement and activity patterns of decapods have been addressed by means of different techniques, such as radiotracking, ultrasonic tracking, and electromagnetic tracking (Vannini & Cannicci, 1995; Smith et al., 2000). These techniques allow for the tracking of pathways followed by individuals, but each has specific problems that limit its applications. For example, radio energy is severely attenuated by seawater, whereas ultrasonic signals are both attenuated and reflected by seabed features and are unsuitable for decapods inhabiting uneven rocky seabed, especially if, as in

Panulirus guttatus, they spend most of their time within shelters (Van der Meeren, 1997; Smith et al., 1998). Electromagnetic tracking, by virtue of its complex array of aerials, has an even shorter range of detection than acoustic tracking (Jernakoff et al., 1987; Wolcott, 1995). In addition, all of these techniques measure activity in spatial scales smaller than the one addressed in our study, which called for an experimental displacement of a large number of individuals to increasingly longer distances from their home site (Vannini & Cannicci, 1995). Once the homing abilities of a species are determined by experimental displacements, the actual pathways followed by individuals can be elucidated by tracking a few individuals.

Because traps fail to attract all the lobsters in their vicinity (Jernakoff & Phillips, 1988), some authors (e.g., Herrnkind, 1980) have discouraged the use of traps for studying the homing and home range of palinurids. However, in complex habitats such as coral reefs, fixed traps help to assess the return of tagged lobsters to their sites of original capture. For example, the first evidence of a homing instinct in tagged and displaced *P. argus* was obtained by means of traps by Creaser & Travis (1950). A more refined characterization of the pathways followed by homing individuals of *P. argus* was later obtained with sonic tags by Herrnkind & McLean (1971) and Herrnkind et al. (1975). When using fixed traps, it is also useful to compare the percentage of recaptures of undisplaced vs. displaced individuals. For example, Chittleborough (1974), using fixed traps, recaptured 12-30% of juveniles of *Panulirus cygnus* that had been released at the same point of capture, but only 2% of individuals that had been displaced 140-400 m from their home patch, which suggested a home range smaller than the range of displacement distances. Later, with electromagnetic devices, Jernakoff et al. (1987) determined a home range of ~150 m in radius for juvenile *P. cygnus*. In a study where individuals of *P. guttatus* were caught in fixed traps, tagged, and released at the very same point of capture, Negrete-Soto et al. (2002) recaptured 23.4% individuals, a value comparable to our overall result of 20.7% of recaptures in traps of displaced individuals. This indicates that *P. guttatus* has homing abilities despite being a sedentary species. Moreover, although traps tend to undersample female *P. guttatus* (cf. Sutcliffe, 1953; Evans et al., 1995; Briones-Fourzán & Contreras-Ortiz, 1999), the similar sex ratios of our captured and recaptured lobsters indicate that individuals of both sexes (including ovigerous females) exhibit similar homing abilities.

A relatively high percentage of the lobsters displaced into the reef lagoon (310°) was recaptured. These lobsters were released at night, which possibly reduced their risk of predation and increased their chances to return to the coral reef. Known predators of *P. guttatus* are triggerfishes, groupers, snappers, and octopuses (Briones-Fourzán, 1991; Negrete-Soto et al., 2002). In general, these lobsters wandered at random during the 7 min of individual observation, and only a few oriented towards the coral reef within this period, but the results suggest

that individual *P. guttatus* displaced to open areas will eventually head back to the coral reef. In a similar experimental displacement, individual crabs (*Thalamita crenata* Latreille, 1829) were released 5, 20, and 50 m away from their holes. Most recaptured animals found their way back from 5 and 20 m, but not from 50 m (Vannini & Cannicci, 1995). However, in the present study, most of the recaptured lobsters from 310° had been displaced 200 m into the reef lagoon, whereas a lower proportion of lobsters displaced 100 and 50 m into the reef lagoon were recaptured. We do not have a clear explanation for this phenomenon.

In contrast, none of the lobsters displaced 200 m in the two directions (20° and 200°) along the axis of the reef tract were recaptured. Two alternative explanations for these results are (a) that these lobsters might have suffered a higher predation rate because they were released in daytime, or (b) that 200 m along the reef tract is beyond the familiar home range of individual *P. guttatus*, hence these lobsters were not able to find their way back to their home patch. We believe that (b) is a more likely explanation because, although on several occasions we observed individual triggerfish (*Balistes vetula* Linnaeus, 1758) stalking or attacking lobsters that had been displaced along the reef tract in daytime, most of these lobsters tended to retreat rapidly into any available crevice, and those that were recaptured were in the same area of their original patch.

Regardless of the time elapsed between subsequent captures, our recaptures of displaced *P. guttatus* were not random, i.e., displaced individuals were apparently not recaptured in the first trap they encountered. On the contrary, all but one of our *P. guttatus* were recaptured in the same area of their original home patch, and most occurred in the same trap or in traps up to 30 m away from the trap of original capture. This suggests that individual *P. guttatus* remain in a more or less defined home range on the reef for a long period, and that their movements in the reef habitat are not purely random (Hazlett & Rittschof, 1975). Rather, the ability of individual *P. guttatus* to relocate their familiar home range suggests a non-random, non-diffusive movement pattern (Vannini & Cannicci, 1995), i.e., true homing (Papi, 1992). Therefore, we conservatively estimate the home range of individual *P. guttatus* to lie within a radius of 100 m along the reef tract (the maximum distance of displacement along the axis of the reef tract from where lobsters returned to their home patch). Within this home range, individuals may typically use more than one shelter, as other decapods do that depend on natural crevices or holes in the substrate for refuge (e.g., *P. argus*, cf. Herrnkind et al., 1975; *Eriphia smithii* MacLeay, 1838, cf. Vannini and Gherardi, 1988; *Homarus americanus* H. Milne Edwards, 1837, cf. Karnofsky et al., 1989; *Jasus edwardsii*, cf. MacDiarmid et al., 1991). In the study of Negrete-Soto et al. (2002), 72 of 74 undisplaced adult *Panulirus guttatus* were recaptured in a 50 m radius from their

point of original capture, which further supports our home range estimate of 100 m in radius for *P. guttatus*.

The homing ability of *P. guttatus* suggests a strong sense of orientation, which has been confirmed in the second stage of the study. Lobsters displaced 500 m into the reef lagoon significantly headed back towards the reef tract, which lay in the same mean direction as the wave surge, the wind, and the ripples in the sand. Because *P. guttatus* does not occur in the reef lagoon, the displacement into this habitat completely removed these individuals from their familiar surroundings. Although more than one stimulus may have been instrumental in the directions taken by these lobsters, wave surge was probably the strongest cue back to the reef tract. The dominant winds in our study area are trade winds that reach the coast from the eastern quadrant. After breaking on the reef, waves reach the coast parallel to the reef tract (Merino & Otero, 1991), and owing to the shallowness of the reef lagoon, wave surge produces clear parallel sand ripples. In contrast, currents within the reef lagoon vary on a daily basis, but tend to run parallel to the coastline, either in a NW or a SE direction (Merino & Otero, 1991). Therefore, current direction could not guide lobsters back to the reef tract. Both magnetic (Lohmann et al., 1995) and upwave orientation have been documented for *P. argus*, and the latter is believed to be a shelter-seeking tactic for lobsters when they are outside their familiar home range (Walton & Herrnkind, 1977; Nevitt et al., 1995). The relatively wide dispersion angle of the headings of our individual *P. guttatus* supports the hypothesis of Herrnkind (1983) that lobsters show menotactic responses to hydrodynamic cues rather than a mere bi-directional rheotaxis to a particular stimulus. Our results indicate that hydrodynamic cues may also be important in the orientation of non-migratory lobster species such as *P. guttatus*.

Once back on the reef tract (a familiar environment), individual *P. guttatus* that were not displaced beyond a threshold distance may have used other stimuli to find their particular home patch. Whether these stimuli were visual, tactile, magnetic, chemical, or a combination of all is yet to be determined. In *Jasus edwardsii*, short-range movements seem to follow the topography of the reef (MacDiarmid et al., 1991). In our study, lobsters displaced into the fore-reef may have used the slope of the reef and/or the reverse wave surge to return to the back-reef area. However, according to Herrnkind (1983), the choice of direction in a familiar area is set at a given internal state by the chemical environment at that location. It has been shown experimentally that individuals of *Panulirus interruptus* (Randall, 1840) (cf. Zimmer-Faust et al., 1985), *P. argus* (cf. Ratchford & Eggleston, 1998), and *Jasus edwardsii* (cf. Butler et al., 1999) are attracted to shelters by odours released by conspecifics already residing in that shelter. This chemosensitivity to water-borne substances, released either by conspecifics or by other components in the substrate,

may also exist in *Panulirus guttatus*. Because all our test individuals were adults (size at first maturity of females: ~38 mm CL; of males: ~48 mm CL, Sharp et al., 1997) and hence had probably remained in their familiar home range for a relatively long time, learning by experience (Schöne, 1961, 1965) may also have played an important role in their ability to relocate their home area. Finally, it is important to mention that our recaptured lobsters backtracked even though they were displaced via a highly indirect course, i.e., transporting them to a location 6 km away from their home patches where they were kept for several hours before being released.

We consider our results as provisional because of our relatively small sample. However, these results show that *P. guttatus*, despite being an obligate reef-dweller as well as a non-migratory and reclusive species, exhibits homing and orientation abilities similar to its more mobile, highly migratory congeners. When displaced to unfamiliar environments, such as the reef lagoon, *P. guttatus* may use wave surge as a cue to return to the coral reef. When displaced along the reef tract, individual *P. guttatus* are able to find their way back to their familiar home range from up to 100 m away. Daily foraging movements and the pathways followed by individuals remain unknown, but may be elucidated in the near future by means of electromagnetic tracking (Jernakoff et al., 1987; Smith et al., 2000).

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